Hunt Feng

University of Saskatchewan | Physics and Engineering Physics, M.Sc
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- GPU Software Engineer with 3+ years in HPC and GPU-accelerated simulations
- Expertise in C++/CUDA/Python, scientific computing and web development
- Skilled in PyTorch, training neural networks including Physics-Informed Neural Nets

Experience

Full-stack Developer
Sophosia Software Inc. | 2021 - present
- Led a team of 3 through ideation to pre-launch using Agile
- Grew a community to 200 users, gained 150+ stars and 4.2k downloads
- Automated releases, saving 30 minutes per update via GitHub Actions CI/CD

Graduate Student, Computational Plasma Physics
University of Saskatchewan | 2022 - 2024
- Set up MHD (Magnetohydrodynamics) simulations on multi-node HPC clusters
- Optimized and benchmark WarpX PIC (Particle-In-Cell) simulations on GPU
- Reduced PIC simulation runtime by ~30% through CUDA kernel optimization
- Reduced 30 minutes per analysis by automating post-processing using Python

Computational Plasma Engineer
General Fusion | 2021 - 2022
- Accelerated MHD simulations by ~30% using time-implicit scheme
- Ensured numerical stability in MHD simulations using PPDG Method (Wu 2018)
- Developed a GPU-accelerated PIC (Particle-in-Cell) code in Taichi Lang for ionization analysis
- Deployed solvers across multiple backends (OpenMP, CUDA, Metal) via Taichi Lang

Research Assistant
SFU's Big Data Hub | 2019
- Built OpenCV application with 93% accuracy for handwritten surveys
- Boosted mineral yield by 21% through neural network pattern recognition

Projects

Semi-Lagrangian Vlasov Solver with Immersed Boundary (GitHub: HuntFeng/vlasolver)
- Enabled GPU-based Vlasov-Poisson solver with immersed boundary method
- Debugged using cuda-gdb (compiled with -g, -O0 flags)
- Profiled with Nsight System / Compute
- Improved performance 10% by resolving thread contention

Physics-Informed Neural Net for Landau Damping (GitHub: HuntFeng/landau_damping_PINN)
- Built a PINN in PyTorch to model Landau damping dynamics
- Trained on only 10% of PIC simulation data to reduce compute costs
- Achieved 99% energy prediction accuracy within first 2 unit times

Presentations

- (Poster, APS DPP 2025) Noise-Free Grid-based Vlasov Modeling of Plasma-Object Interactions via a Unified Ghost-Fluid Immersed-Boundary Method.
- (Poster, APS DPP 2023) *Stability of the Transonic Plasma Flow in the Magnetic Nozzle.*
- (Poster, CUPC 2021) *GPU-based Provably Positive Discontinuous Galerkin method for Multidimensional Ideal Magnetohydrodynamics.*

Skills

Programming: Python (Scientific packages, PyTorch, ...), C/C++, CUDA, HIP (transferable from CUDA), Tachi Lang, Kokkos, Matlab, Lua, MPI/OpenMP, Typescript, Vue.js, (No)SQL
DevOps & Tools: Slurm, Docker, Apptainer, CMake, Nsight System/Compute, GDB, Git(Hub)
HPC/Cloud Platforms: Niagara, Narval, Mist, Plato, GCP, AWS EC2
Simulation Tools: WarpX, Bout++, EDIPIC, Fusion 360, Paraview
Knowledge: Numerical analysis, Sparse matrix, PIC, MHD, Machine Learning, PINN, Optimization